



Fisher-Orthogonal Parameter Manifolds: Enforcing Hard Constraints in Hypernetwork-based Continual Learning



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Introduction

Continual learning asks whether a neural network can learn a sequence of tasks without forgetting what it learned before. In practice, networks fail catastrophically: gradient descent on a new task overwrites weights that encoded prior knowledge, collapsing old-task accuracy within just a few update steps.

The field has produced three families of fixes — **regularization** (penalize changes to important weights), **replay** (store and re-train on old examples), and **architecture** methods (partition capacity per task) — but none fully solve the problem in a principled, parameter-efficient way. Most methods fall short because they operate in Euclidean space (e.g., OGD; Farajtabar et al., 2020), which fails to capture how a distribution-producing machine actually functions. By ignoring the sensitivity of the output to weight changes, these methods rely on an unreliable notion of importance. To fix this, I propose exploring second-derivative regularizers that respect the true geometry of the parameter space.

FOPNG: Fisher-Orthogonal Projected Natural Gradient Descent

Fisher Information and Natural Gradient

The **Fisher information matrix** $\mathcal{F}_\theta$ measures how sensitively the model's output distribution responds to each parameter:

$$\mathcal{F}_\theta = \mathbb{E}[\nabla_\theta \log p(y|x; \theta) \nabla_\theta \log p(y|x; \theta)^\top] \quad (1)$$

Standard gradient descent ignores this curvature. **Natural gradient descent** (Amari, 1998) corrects it by preconditioning with $\mathcal{F}_\theta^{-1}$, yielding updates that are equal-sized in distribution space and reparameterization-invariant:

$$\Delta\theta_{\text{nat}} = -\eta \mathcal{F}_\theta^{-1} \nabla_\theta L \quad (2)$$

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Orthogonal Projection

After each task k , the gradient g_k is stored in a memory matrix $G = [g_1 | \dots | g_{t-1}]$. New updates are projected onto the **Fisher-orthogonal complement** of G — directions that cannot alter prior task outputs:

$$\Delta\theta = -\eta (I - G(G^\top \mathcal{F}_\theta G)^{-1} G^\top \mathcal{F}_\theta) \mathcal{F}_\theta^{-1} \nabla_\theta L_t \quad (3)$$

The term $G(G^\top \mathcal{F}_\theta G)^{-1} G^\top \mathcal{F}_\theta$ is a projection that reads off and removes any component of the new gradient that would disturb old tasks. What remains is a geometrically hard-constrained update.

Combining those two ideas gives us **FOPNG** as proposed by Garg et al. (2026).

Hypernetworks

A **hypernetwork** $h(\cdot; \phi)$ generates the weights of a target network rather than making predictions itself. Each task t receives a learned embedding z_t , producing task-specific weights on the fly:

$$\theta_t = h(z_t; \phi) \quad (4)$$

The shared parameters ϕ do not grow with the number of tasks, and each task gets a fully specialized network by varying z_t . This approach was first applied to continual learning by von Oswald et al. (2020). Applying FOPNG to ϕ means the Fisher matrix and gradient memory are maintained only for this compact set of parameters, not the larger target network.

For very large target networks, **chunking** (Ha et al., 2017) keeps ϕ tractable: the hypernetwork generates target weights in small blocks, each conditioned on z_t and a chunk index, rather than all at once.

Baselines

SGD and **Adam** are naive baselines with no forgetting protection, establishing the lower bound.

EWC (Elastic Weight Consolidation; Kirkpatrick et al., 2017) penalizes changes to weights deemed important for prior tasks via the diagonal Fisher, but operates in Euclidean space.

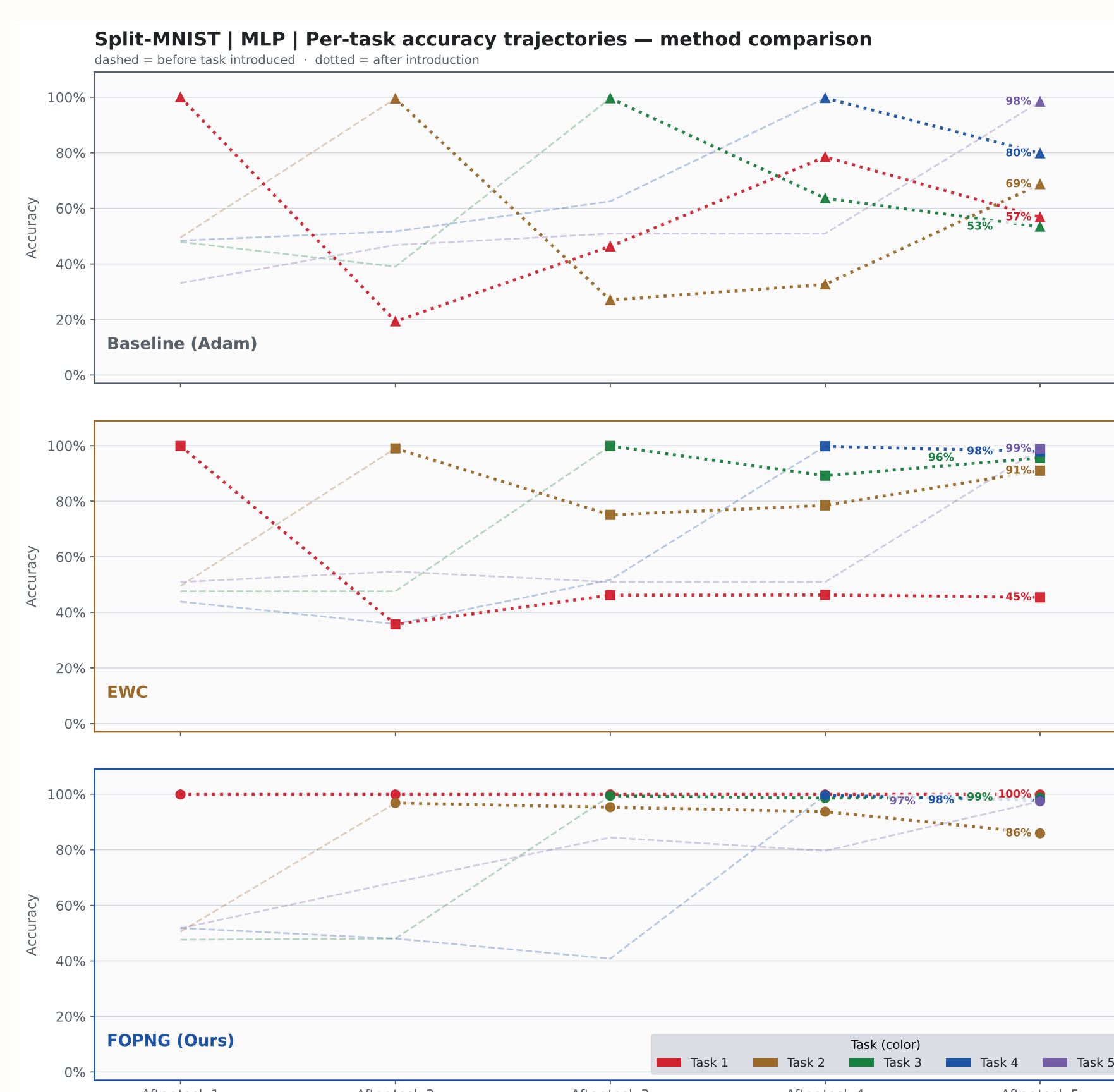


Figure 1: First preliminary results for Split-MNIST. Standard MLP performance comparison. Notice the severe performance degradation in Baseline (Adam) and partial forgetting in EWC.

OGD (Orthogonal Gradient Descent; Farajtabar et al., 2020) projects gradients onto the Euclidean orthogonal complement of prior gradients — the closest Euclidean analogue to FOPNG.

FNG (Fisher Natural Gradient) applies natural gradient updates without orthogonal projection, isolating the contribution of Fisher preconditioning alone.

Benchmarks

Split-MNIST splits MNIST into 5 binary tasks (e.g. 0 vs 1, then 2 vs 3). The standard entry-level benchmark for sequential learning.

Permuted-MNIST applies a different random pixel permutation per task, all sharing the same label space. Tests long-horizon retention over 10+ tasks.

Split-CIFAR-10 and **Split-CIFAR-100** apply the split protocol to natural images: 5 tasks over 10 classes, and 20 tasks over 100 classes respectively. Higher visual complexity and more tasks make these substantially harder.

Evaluation Metrics

I will report **average accuracy** across all tasks after training, **backward transfer** (how much new training degrades old tasks), **forward transfer** (whether prior tasks help new ones), and **per-task accuracy trajectories** to visualize forgetting dynamics over time. I will also introduce many tangential metrics for deeper analysis.

Summary and Conclusions

This project aims to verify a novel continual learning framework combining FOPNG with Hypernetworks. I aim to show that enforcing orthogonality within the Fisher-Riemannian manifold, rather than standard Euclidean space, leverages essential second-derivative information to more effectively preserve task-specific knowledge. Through benchmarks like Split-MNIST, I will test whether this geometrically-aware approach consistently outperforms Euclidean-based baselines like OGD.

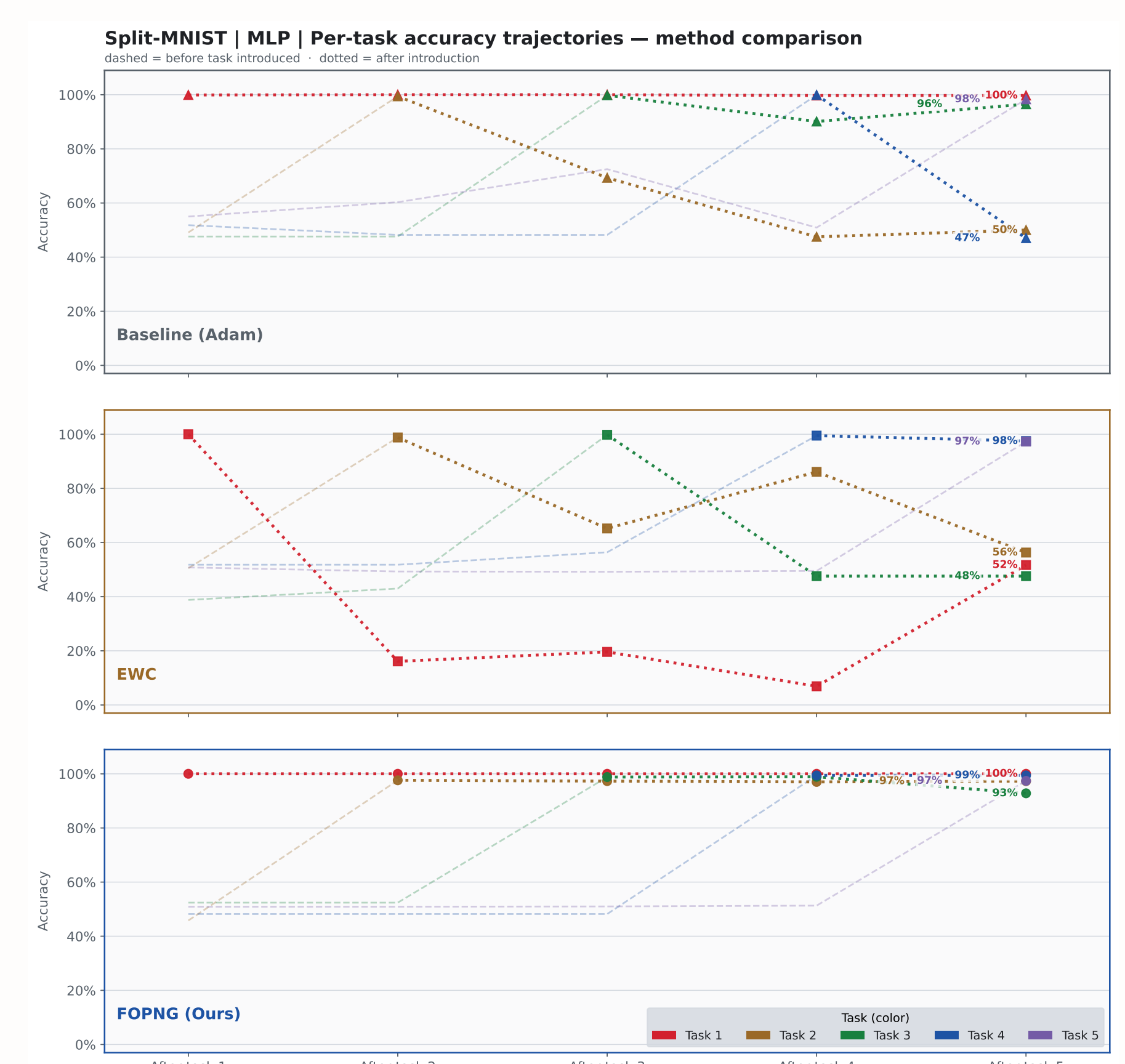


Figure 2: Full (not chunked) Hypernetwork performance on Split-MNIST. FOPNG successfully maintains high accuracy across all sequential tasks without catastrophic forgetting.